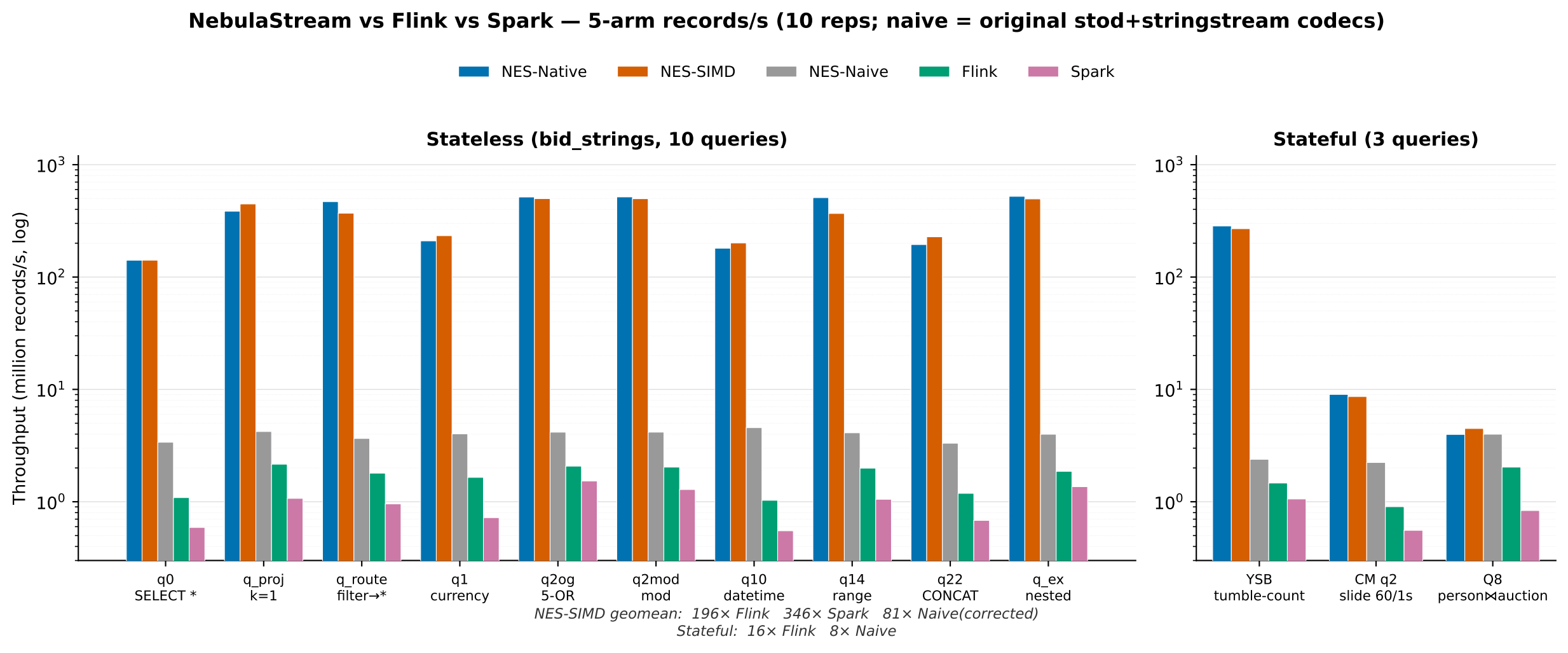
NebulaStream vs Flink vs Spark — 5-arm records/s (10 reps; naive = original stod+stringstream codecs)
NES-Native
NES-SIMD
NES-Naive
Flink
Spark
Stateless (bid_strings, 10 queries)
Stateful (3 queries)
Throughput (million records/s, log)
10^3
10^2
10^1
10^0
q0 SELECT *
q_proj k=1
q_route filter→*
q1 currency
q2og 5-OR
q2mod mod
q10 datetime
q14 range
q22 CONCAT
q_ex nested
YSB tumble-count
CM q2 slide 60/1s
Q8 person⋈auction
NES-SIMD geomean: 196× Flink 346× Spark 81× Naive(corrected)
Stateful: 16× Flink 8× Naive